risk_per_100k_by_state.ipynb

In [1]

```
import os
from pathlib import Path

import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
import seaborn as sns
from dotenv import load_dotenv
from supabase import create_client

%matplotlib inline

project_root = Path.cwd().parent
env_path = project_root / ".env"

load_dotenv(dotenv_path=env_path, override=True)

SUPABASE_URL = os.getenv("SUPABASE_URL")
SUPABASE_KEY = os.getenv("SUPABASE_KEY")

supabase = create_client(SUPABASE_URL, SUPABASE_KEY)
```

In [2]

```
page_size = 1000
all_rows = []
start = 0

while True:
    response = (
        supabase
        .table("incident")
        .select("*")
        .order("Incident_ID")
        .range(start, start + page_size - 1)
        .execute()
    )

    data = response.data
    if not data:
        break

    all_rows.extend(data)
    start += page_size

incident_df = pd.DataFrame(all_rows)
incident_df["Number_Victims"] = pd.to_numeric(incident_df["Number_Victims"],
errors="coerce")
```

In [3]

```
page_size = 1000
all_rows = []
start = 0

while True:
    response = (
        supabase
        .table("shooter")
        .select("*")
        .order("Incident_ID")
        .range(start, start + page_size - 1)
        .execute()
    )

    data = response.data
    if not data:
        break

    all_rows.extend(data)
    start += page_size

shooter_df = pd.DataFrame(all_rows)
```

In [4]

```
page_size = 1000
all_rows = []
start = 0

while True:
    response = (
        supabase
        .table("victim")
        .select("*")
        .order("Incident_ID")
        .range(start, start + page_size - 1)
        .execute()
    )

    data = response.data
    if not data:
        break

    all_rows.extend(data)
    start += page_size

victim_df = pd.DataFrame(all_rows)
```

In [5]

```
page_size = 1000
all_rows = []
start = 0

while True:
    response = (
        supabase
        .table("weapon")
        .select("*")
        .order("Incident_ID")
        .range(start, start + page_size - 1)
        .execute()
    )

    data = response.data
    if not data:
        break

    all_rows.extend(data)
    start += page_size

weapon_df = pd.DataFrame(all_rows)
```

In [6]

```
all_rows = []
start = 0
batch_size = 1000

while True:
    response = (
        supabase
        .table("enrollment_state_year_mat")
        .select("*")
        .range(start, start + batch_size - 1)
        .execute()
    )

    data = response.data

    if not data:
        break

    all_rows.extend(data)
    start += batch_size

enrollment_df = pd.DataFrame(all_rows)
```

In [7]

```
valid_states = {
    "AL", "AK", "AZ", "AR", "CA", "CO", "CT", "DE", "FL", "GA",
    "HI", "ID", "IL", "IN", "IA", "KS", "KY", "LA", "ME", "MD",
    "MA", "MI", "MN", "MS", "MO", "MT", "NE", "NV", "NH", "NJ",
    "NM", "NY", "NC", "ND", "OH", "OK", "OR", "PA", "RI", "SC",
    "SD", "TN", "TX", "UT", "VT", "VA", "WA", "WV", "WI", "WY", "DC"
}

incident_df_from_1987 = incident_df[incident_df["State"].isin(valid_states)].copy()
incident_df_from_1987["Year"] = pd.to_numeric(incident_df_from_1987["Year"],
errors="coerce")
incident_df_from_1987["Victims_Killed"] =
pd.to_numeric(incident_df_from_1987["Victims_Killed"], errors="coerce").fillna(0)
incident_df_from_1987 = incident_df_from_1987.dropna(subset=["Year"]).copy()
incident_df_from_1987["Year"] = incident_df_from_1987["Year"].astype(int)
incident_df_from_1987 = incident_df_from_1987[incident_df_from_1987["Year"] >=
1987].copy()

incident_agg = (
    incident_df_from_1987
    .groupby(["State", "Year"], as_index=False)
    .size()
    .rename(columns={"size": "incident_count"})
)
```

In [8]

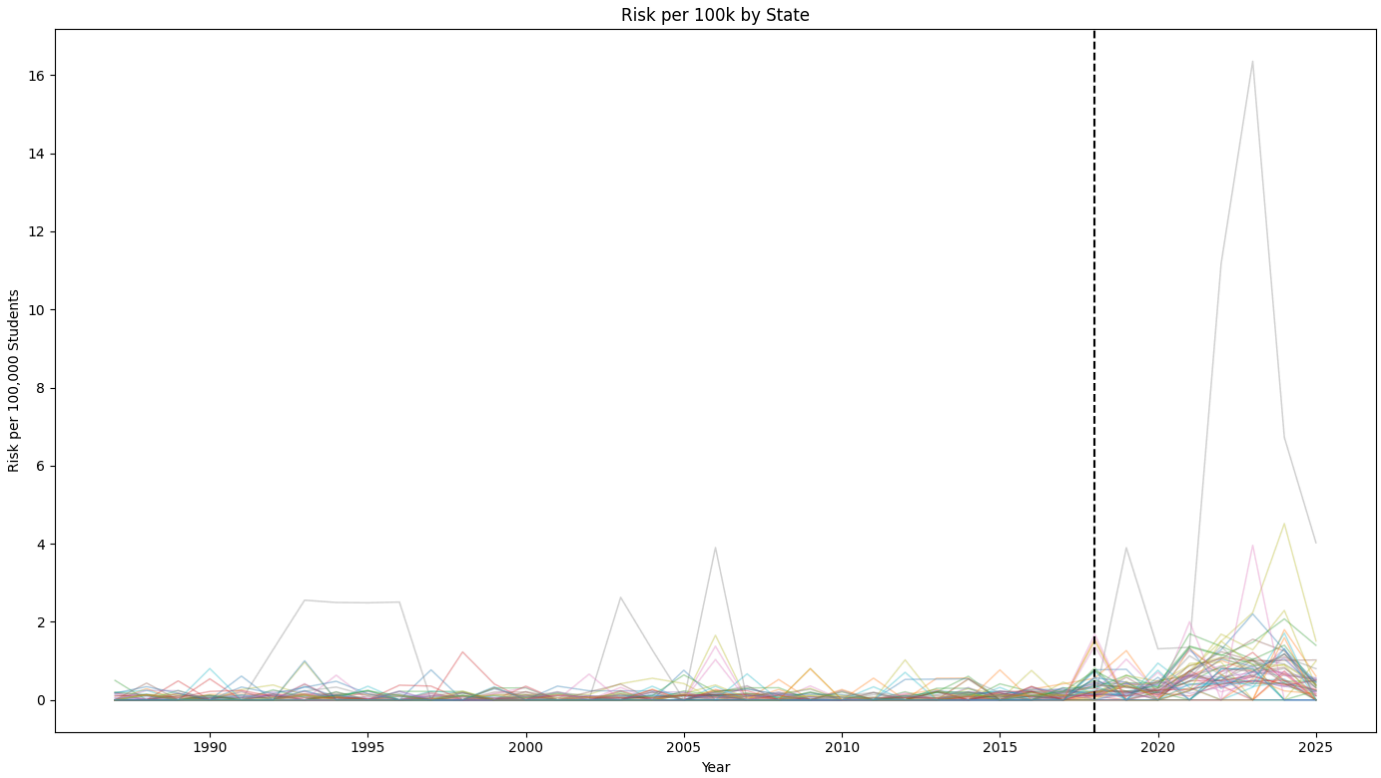
```
enrollment_df["state"] = enrollment_df["state"].astype(str).str.upper()

state_abbrev = {
    "ALABAMA": "AL", "ALASKA": "AK", "ARIZONA": "AZ", "ARKANSAS": "AR",
    "CALIFORNIA": "CA", "COLORADO": "CO", "CONNECTICUT": "CT",
    "DELAWARE": "DE", "DISTRICT OF COLUMBIA": "DC", "FLORIDA": "FL",
    "GEORGIA": "GA", "HAWAII": "HI", "IDAHO": "ID", "ILLINOIS": "IL",
    "INDIANA": "IN", "IOWA": "IA", "KANSAS": "KS", "KENTUCKY": "KY",
    "LOUISIANA": "LA", "MAINE": "ME", "MARYLAND": "MD",
    "MASSACHUSETTS": "MA", "MICHIGAN": "MI", "MINNESOTA": "MN",
    "MISSISSIPPI": "MS", "MISSOURI": "MO", "MONTANA": "MT",
    "NEBRASKA": "NE", "NEVADA": "NV", "NEW HAMPSHIRE": "NH",
    "NEW JERSEY": "NJ", "NEW MEXICO": "NM", "NEW YORK": "NY",
    "NORTH CAROLINA": "NC", "NORTH DAKOTA": "ND", "OHIO": "OH",
    "OKLAHOMA": "OK", "OREGON": "OR", "PENNSYLVANIA": "PA",
    "RHODE ISLAND": "RI", "SOUTH CAROLINA": "SC",
    "SOUTH DAKOTA": "SD", "TENNESSEE": "TN", "TEXAS": "TX",
    "UTAH": "UT", "VERMONT": "VT", "VIRGINIA": "VA",
    "WASHINGTON": "WA", "WEST VIRGINIA": "WV",
    "WISCONSIN": "WI", "WYOMING": "WY"
}

enrollment_df["State"] = enrollment_df["state"].map(state_abbrev)
enrollment_df = enrollment_df.rename(columns={"year": "Year"})
enrollment_df["Year"] = pd.to_numeric(enrollment_df["Year"], errors="coerce")
enrollment_df["total_students"] = pd.to_numeric(enrollment_df["total_students"],
errors="coerce")
enrollment_df = enrollment_df.dropna(subset=["State", "Year",
"total_students"]).copy()
enrollment_df["Year"] = enrollment_df["Year"].astype(int)
enrollment_df = enrollment_df[enrollment_df["Year"] >= 1987].copy()
enrollment_df = (
    enrollment_df
    .sort_values("total_students", ascending=False)
    .drop_duplicates(["State", "Year"])
)
```

In [9]

```
merged_df_table = enrollment_df.merge(
    incident_agg,
    on=["State", "Year"],
    how="left"
)

merged_df_table["incident_count"] = merged_df_table["incident_count"].fillna(0)
merged_df_table["incident_rate_per_100k"] = np.where(
    merged_df_table["total_students"] > 0,
    (merged_df_table["incident_count"] / merged_df_table["total_students"]) * 100000,
    np.nan
)
merged_df_table["risk_per_100k"] = merged_df_table["incident_rate_per_100k"]
merged_df_table = merged_df_table.drop(columns=["state"])
merged_df_table = merged_df_table.sort_values(["State",
"Year"]).reset_index(drop=True)
```

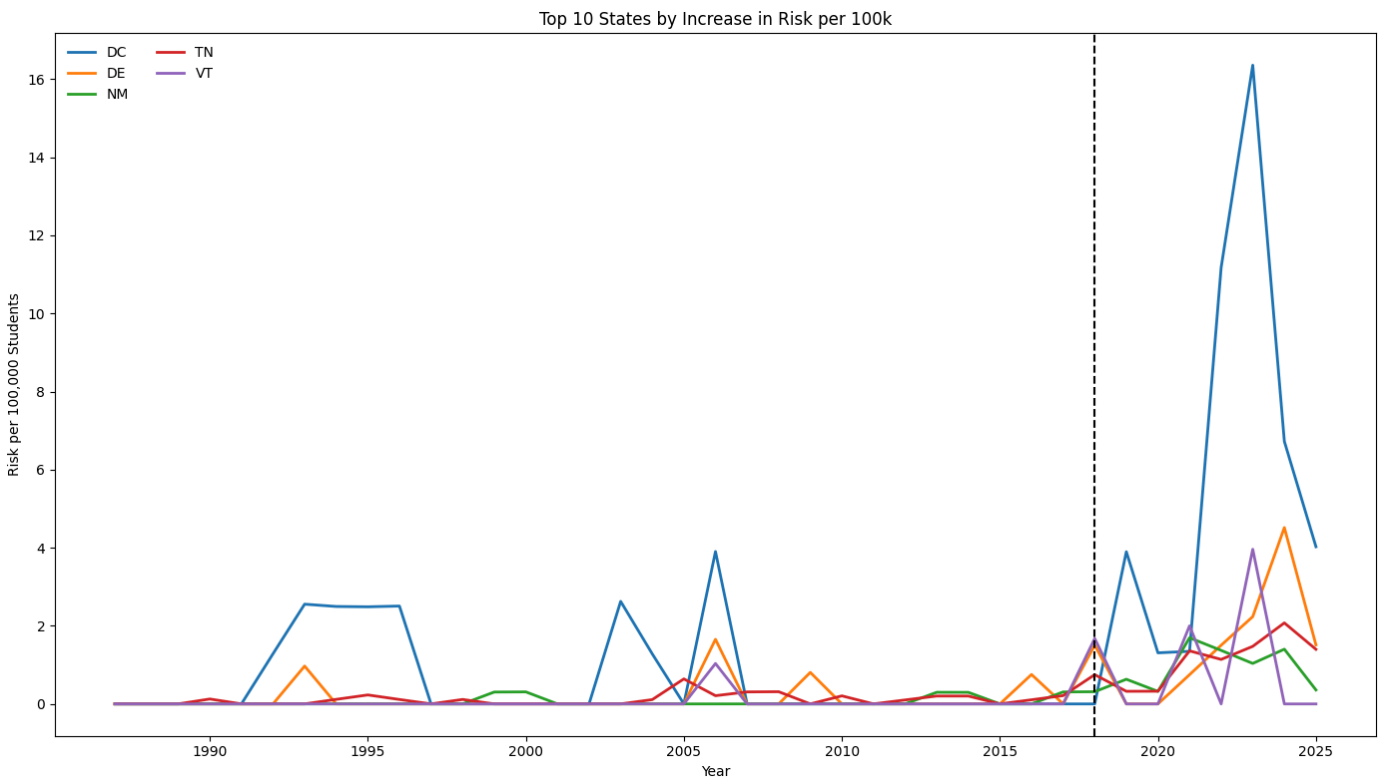
In [10]

```
plt.figure(figsize=(14, 8))
for state, state_df in merged_df_table.groupby("State"):
    plt.plot(state_df["Year"], state_df["risk_per_100k"], alpha=0.35, linewidth=1)

plt.axvline(2018, color="black", linestyle="--", linewidth=1.5)
plt.xlabel("Year")
plt.ylabel("Risk per 100,000 Students")
plt.title("Risk per 100k by State")
plt.tight_layout()
plt.show()
```



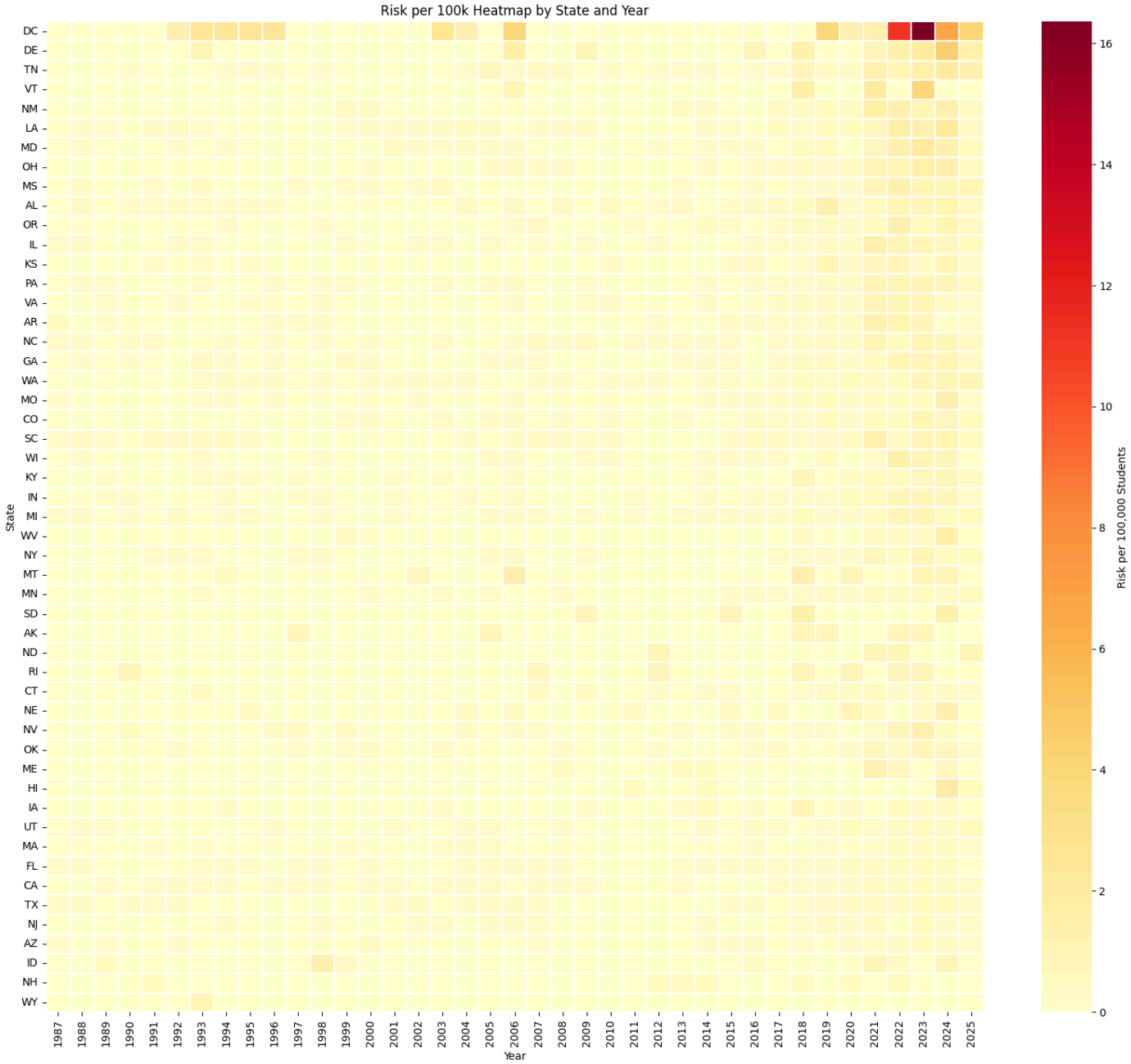
In [11]

```
state_change = (
    merged_df_table
    .assign(period=np.where(merged_df_table["Year"] >= 2018, "post_2018",
"pre_2018"))
    .groupby(["State", "period"])["risk_per_100k"]
    .mean()
    .unstack(fill_value=0)
)
state_change["increase"] = state_change.get("post_2018", 0) -
state_change.get("pre_2018", 0)
top_states = state_change.sort_values("increase", ascending=False).head(5).index
focus_df = merged_df_table[merged_df_table["State"].isin(top_states)].copy()

plt.figure(figsize=(14, 8))
for state, state_df in focus_df.groupby("State"):
    plt.plot(state_df["Year"], state_df["risk_per_100k"], linewidth=2, label=state)

plt.axvline(2018, color="black", linestyle="--", linewidth=1.5)
plt.xlabel("Year")
plt.ylabel("Risk per 100,000 Students")
plt.title("Top 10 States by Increase in Risk per 100k")
plt.legend(ncol=2, frameon=False)
plt.tight_layout()
plt.show()
```



In [12]

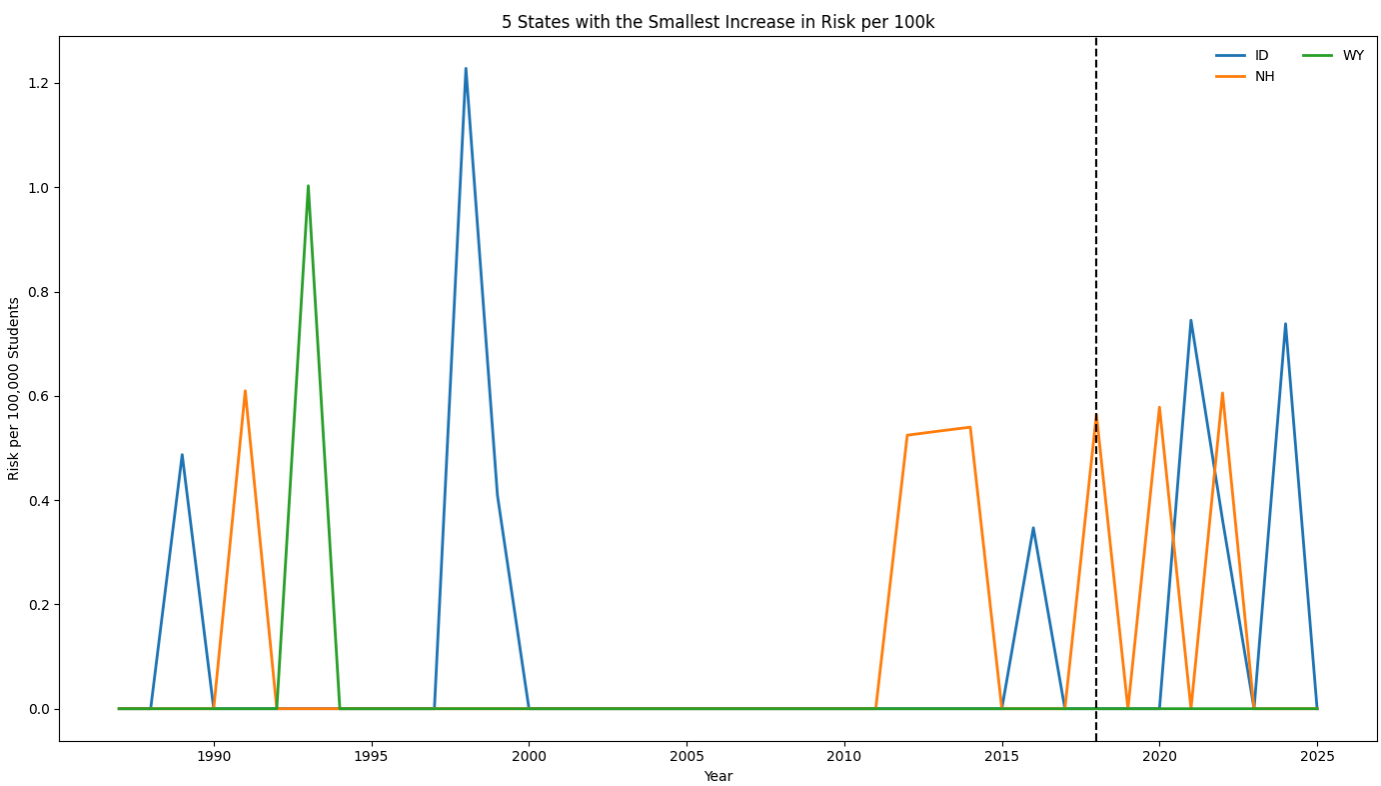
```
state_order = state_change.sort_values("increase", ascending=False).index
heatmap_df = (
    merged_df_table
    .pivot(index="State", columns="Year", values="risk_per_100k")
    .reindex(state_order)
    .fillna(0)
)

plt.figure(figsize=(16, 14))
sns.heatmap(
    heatmap_df,
    cmap="YlOrRd",
    linewidths=0.1,
    linecolor="white",
    cbar_kws={"label": "Risk per 100,000 Students"}
)
plt.xlabel("Year")
plt.ylabel("State")
plt.title("Risk per 100k Heatmap by State and Year")
plt.tight_layout()
plt.show()
```



smallest increase states

In [13]

```
state_change = (
    merged_df_table
    .assign(period=np.where(merged_df_table["Year"] >= 2018, "post_2018",
"pre_2018"))
    .groupby(["State", "period"])["risk_per_100k"]
    .mean()
    .unstack(fill_value=0)
)

state_change["increase"] = state_change.get("post_2018", 0) -
state_change.get("pre_2018", 0)
lowest_increase_states = state_change.sort_values("increase",
ascending=True).head(3).index
focus_df =
merged_df_table[merged_df_table["State"].isin(lowest_increase_states)].copy()

plt.figure(figsize=(14, 8))
for state, state_df in focus_df.groupby("State"):
    plt.plot(state_df["Year"], state_df["risk_per_100k"], linewidth=2, label=state)

plt.axvline(2018, color="black", linestyle="--", linewidth=1.5)
plt.xlabel("Year")
plt.ylabel("Risk per 100,000 Students")
plt.title("5 States with the Smallest Increase in Risk per 100k")
plt.legend(ncol=2, frameon=False)
plt.tight_layout()
plt.show()
```



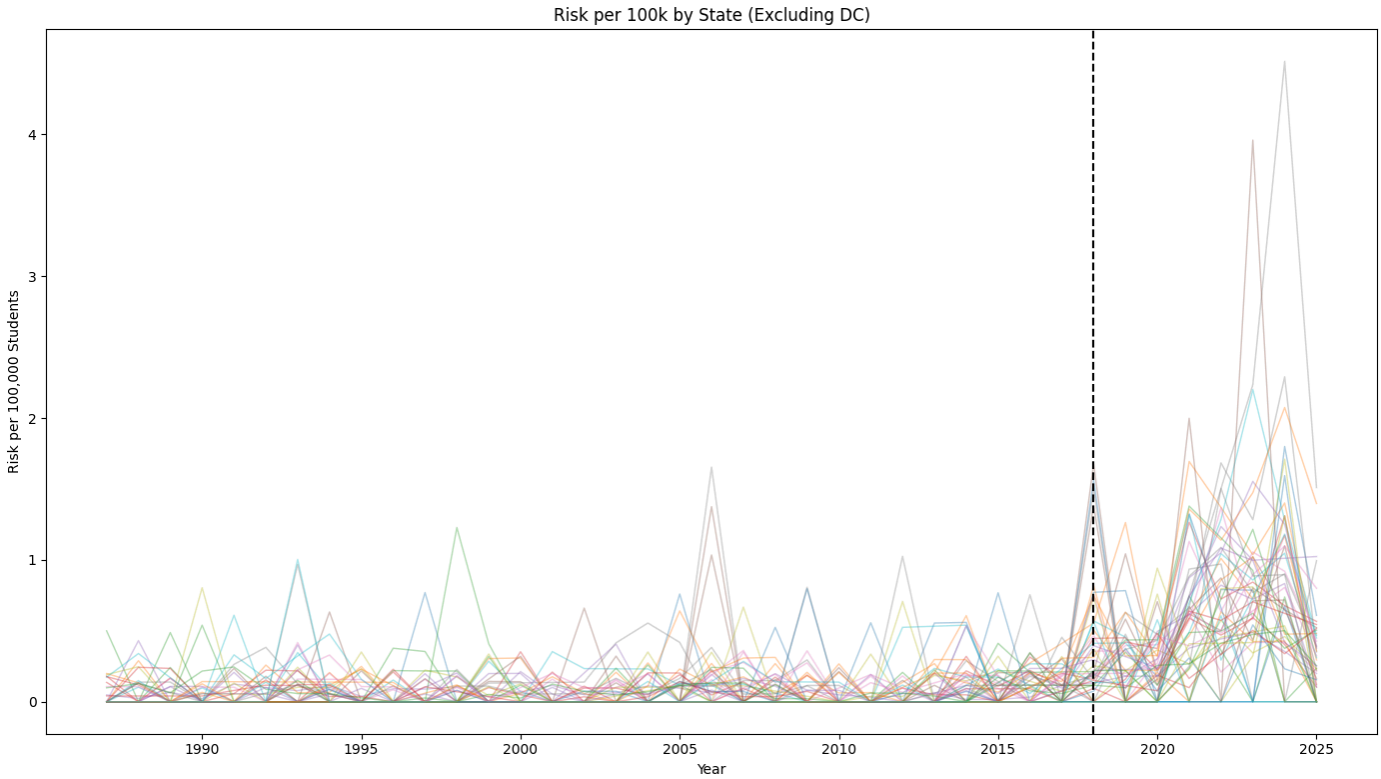
In [14]

```
if "k12_settings_law" not in merged_df_table.columns:
    all_rows = []
    start = 0
    batch_size = 1000

    while True:
        response = (
            supabase
            .table("k12_settings_binary")
            .select("*")
            .range(start, start + batch_size - 1)
            .execute()
        )

        data = response.data
        if not data:
            break

        all_rows.extend(data)
        start += batch_size

    k12_df = pd.DataFrame(all_rows)

    merged_df_table = merged_df_table.merge(
        k12_df,
        left_on=["State", "Year"],
        right_on=["state", "year"],
        how="left"
    ).drop(columns=["state", "year"])

    merged_df_table["k12_settings_law"] =
merged_df_table["k12_settings_law"].fillna(0)

implementation_years = (
    merged_df_table[
        (merged_df_table["State"].isin(lowest_increase_states)) &
        (merged_df_table["k12_settings_law"] == 1)
    ]
    .groupby("State", as_index=False)["Year"]
    .min()
    .rename(columns={"Year": "k12_settings_implementation_year"})
)

implementation_years = (
    pd.DataFrame({"State": list(lowest_increase_states)})
    .merge(implementation_years, on="State", how="left")
)

implementation_years["k12_settings_implementation_year"] = (
    implementation_years["k12_settings_implementation_year"]
    .fillna("Never adopted in panel")
)

implementation_years
```

	State	k12_settings_implementation_year
0	WY	2019.0
1	NH	Never adopted in panel
2	ID	Never adopted in panel

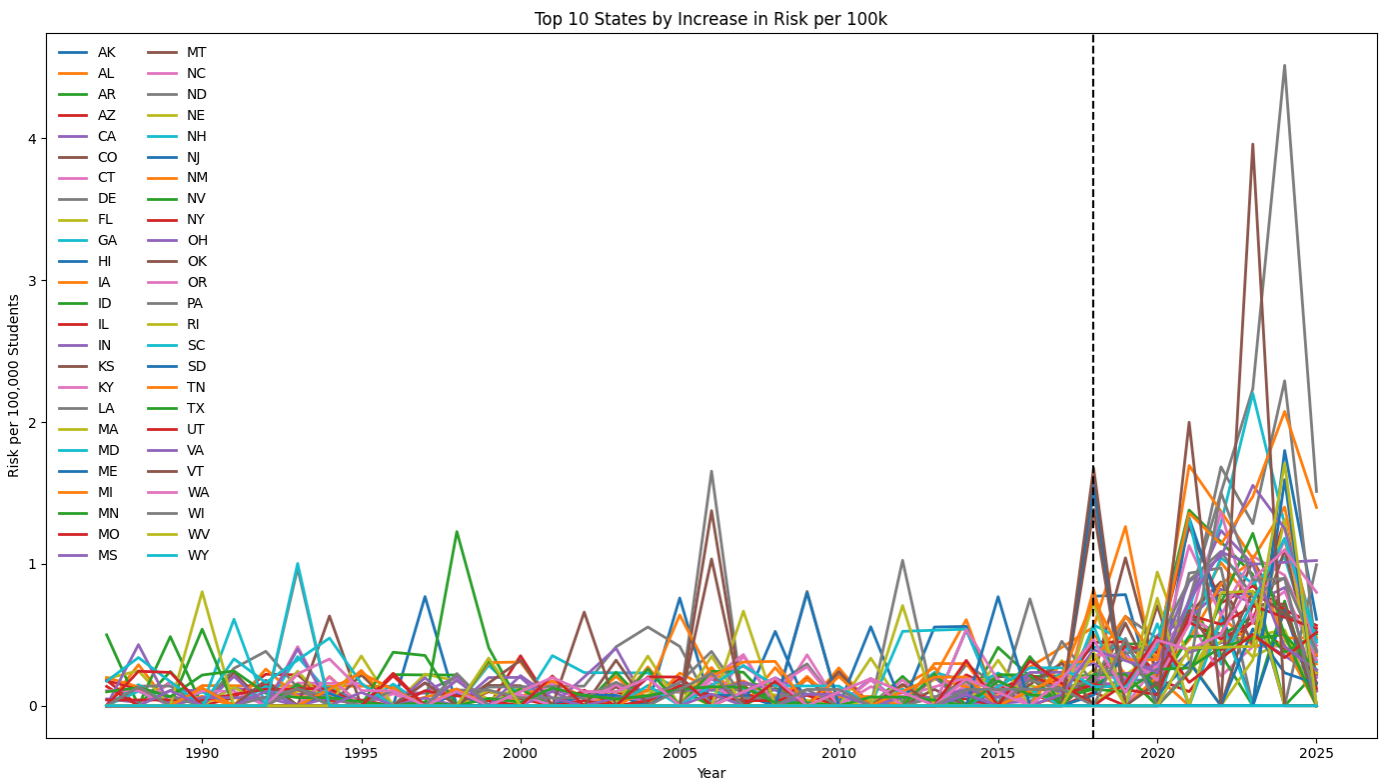
In [15]

```
plt.figure(figsize=(14, 8))
for state, state_df in merged_df_table[merged_df_table["State"] !=
"DC"].groupby("State"):
    plt.plot(state_df["Year"], state_df["risk_per_100k"], alpha=0.35, linewidth=1)

plt.axvline(2018, color="black", linestyle="--", linewidth=1.5)
plt.xlabel("Year")
plt.ylabel("Risk per 100,000 Students")
plt.title("Risk per 100k by State (Excluding DC)")
plt.tight_layout()
plt.show()
```



In [16]

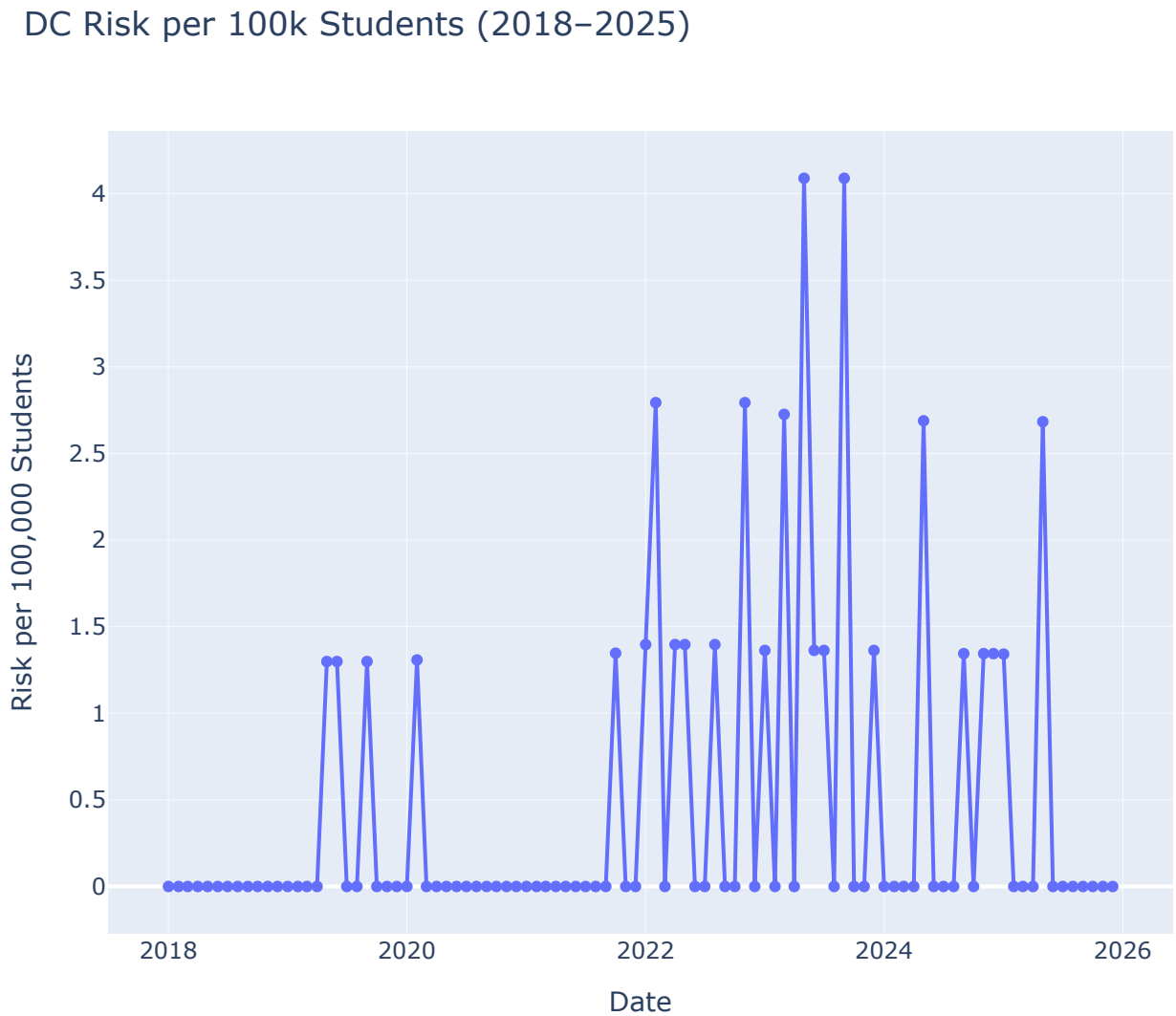
```
state_change = (
    merged_df_table
    .assign(period=np.where(merged_df_table["Year"] >= 2018, "post_2018",
"pre_2018"))
    .groupby(["State", "period"])["risk_per_100k"]
    .mean()
    .unstack(fill_value=0)
)
state_change["increase"] = state_change.get("post_2018", 0) -
state_change.get("pre_2018", 0)
top_states = state_change.sort_values("increase", ascending=False).head(10).index
focus_df = merged_df_table[merged_df_table["State"].isin(top_states)].copy()

plt.figure(figsize=(14, 8))
for state, state_df in merged_df_table[merged_df_table["State"] !=
"DC"].groupby("State"):
    plt.plot(state_df["Year"], state_df["risk_per_100k"], linewidth=2, label=state)

plt.axvline(2018, color="black", linestyle="--", linewidth=1.5)
plt.xlabel("Year")
plt.ylabel("Risk per 100,000 Students")
plt.title("Top 10 States by Increase in Risk per 100k")
plt.legend(ncol=2, frameon=False)
plt.tight_layout()
plt.show()
```



In [17]

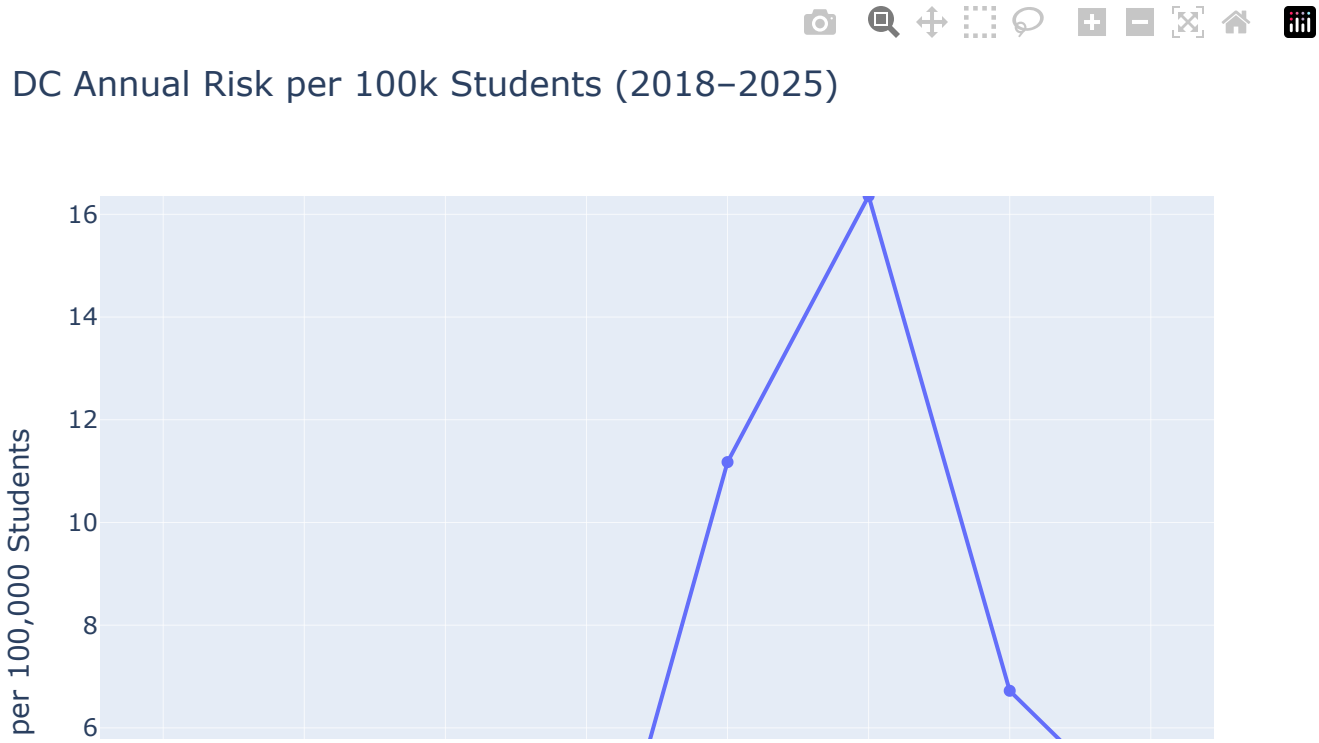
```
import plotly.express as px

dc_enrollment = (
    enrollment_df.loc[
        (enrollment_df["State"] == "DC") &
        (enrollment_df["Year"].between(2018, 2025)),
        ["State", "Year", "total_students"],
    ]
    .rename(columns={"State": "state", "Year": "year", "total_students":
"enrollment"})
    .drop_duplicates(["state", "year"])
)

dc_incidents = incident_df_from_1987.loc[
    (incident_df_from_1987["State"] == "DC") &
    (incident_df_from_1987["Year"].between(2018, 2025)),
    ["State", "Year", "Month", "Incident_ID"],
].copy()

dc_incidents["Month"] = pd.to_numeric(dc_incidents["Month"], errors="coerce")
dc_incidents = dc_incidents.dropna(subset=["Month"])
dc_incidents["Month"] = dc_incidents["Month"].astype(int)

merged_df_table_monthly = (
    pd.MultiIndex.from_product(
        [sorted(dc_enrollment["year"].unique()), range(1, 13)],
        names=["year", "month"],
    )
    .to_frame(index=False)
    .assign(state="DC")
    .merge(
        dc_incidents
        .groupby(["State", "Year", "Month"], as_index=False)
        .agg(incident_count=("Incident_ID", "count"))
        .rename(columns={"State": "state", "Year": "year", "Month": "month"}),
        on=["state", "year", "month"],
        how="left",
    )
    .merge(dc_enrollment, on=["state", "year"], how="left")
)

merged_df_table_monthly["incident_count"] =
merged_df_table_monthly["incident_count"].fillna(0)

dc_monthly = (
    merged_df_table_monthly.loc[
        (merged_df_table_monthly["state"] == "DC") &
        (merged_df_table_monthly["year"].between(2018, 2025))
    ]
    .groupby(["year", "month"], as_index=False)
    .agg(
        total_incidents=("incident_count", "sum"),
        total_enrollment=("enrollment", "sum"),
    )
)

dc_monthly["risk_per_100k"] = np.where(
    dc_monthly["total_enrollment"] > 0,
    (dc_monthly["total_incidents"] / dc_monthly["total_enrollment"]) * 100000,
    np.nan,
)
```

```
dc_monthly["datetime"] = pd.to_datetime(
    {
        "year": dc_monthly["year"],
        "month": dc_monthly["month"],
        "day": 1,
    },
    errors="coerce",
)

dc_monthly = dc_monthly.dropna(subset=["datetime"]).sort_values("datetime")

fig = px.line(
    dc_monthly,
    x="datetime",
    y="risk_per_100k",
    markers=True,
    title="DC Risk per 100k Students (2018–2025)",
    labels={
        "datetime": "Date",
        "risk_per_100k": "Risk per 100,000 Students",
    },
)

fig.update_layout(xaxis_title="Date", yaxis_title="Risk per 100,000 Students")
fig.show()
```



In [18]

```
dc_annual = (
    merged_df_table.loc[
        (merged_df_table["State"] == "DC") &
        (merged_df_table["Year"].between(2018, 2025)),
        ["State", "Year", "incident_count", "total_students", "risk_per_100k"],
    ]
    .copy()
)

dc_annual["incident_count"] = pd.to_numeric(dc_annual["incident_count"],
errors="coerce")
dc_annual["total_students"] = pd.to_numeric(dc_annual["total_students"],
errors="coerce")
dc_annual["risk_per_100k"] = np.where(
    dc_annual["total_students"] > 0,
    (dc_annual["incident_count"] / dc_annual["total_students"]) * 100000,
    np.nan,
)
dc_annual = dc_annual.sort_values("Year")

fig = px.line(
    dc_annual,
    x="Year",
    y="risk_per_100k",
    markers=True,
    title="DC Annual Risk per 100k Students (2018–2025)",
    labels={
        "Year": "Year",
        "risk_per_100k": "Risk per 100,000 Students",
    },
)

fig.update_xaxes(tickmode="array", tickvals=dc_annual["Year"].tolist())
fig.update_yaxes(range=[0, merged_df_table["risk_per_100k"].max()])
fig.update_layout(xaxis_title="Year", yaxis_title="Risk per 100,000 Students")
fig.show()
```



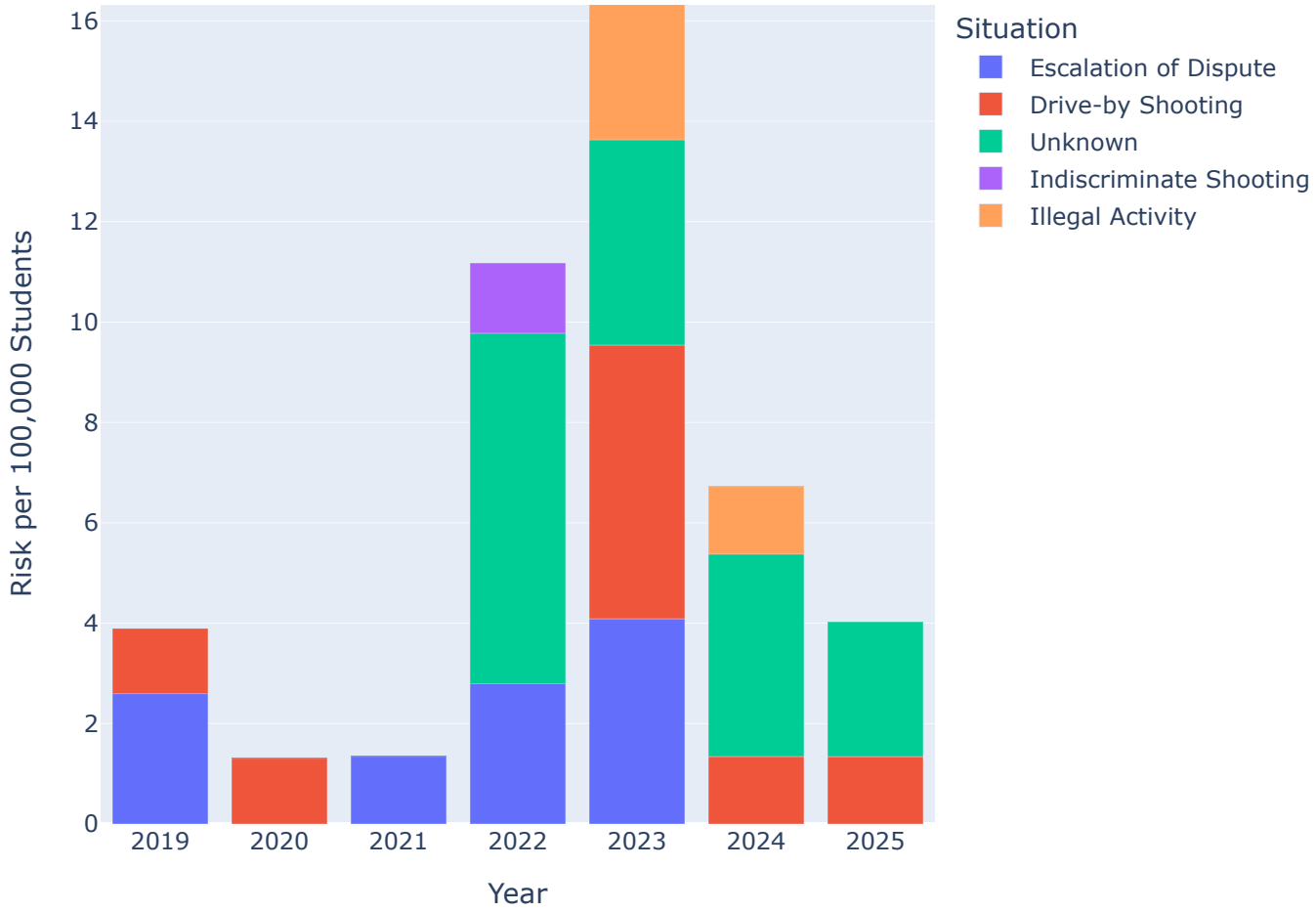
Risk
0
2
4
2018
2019
2020
2021
2022
2023
2024
2025
Year

In [19]

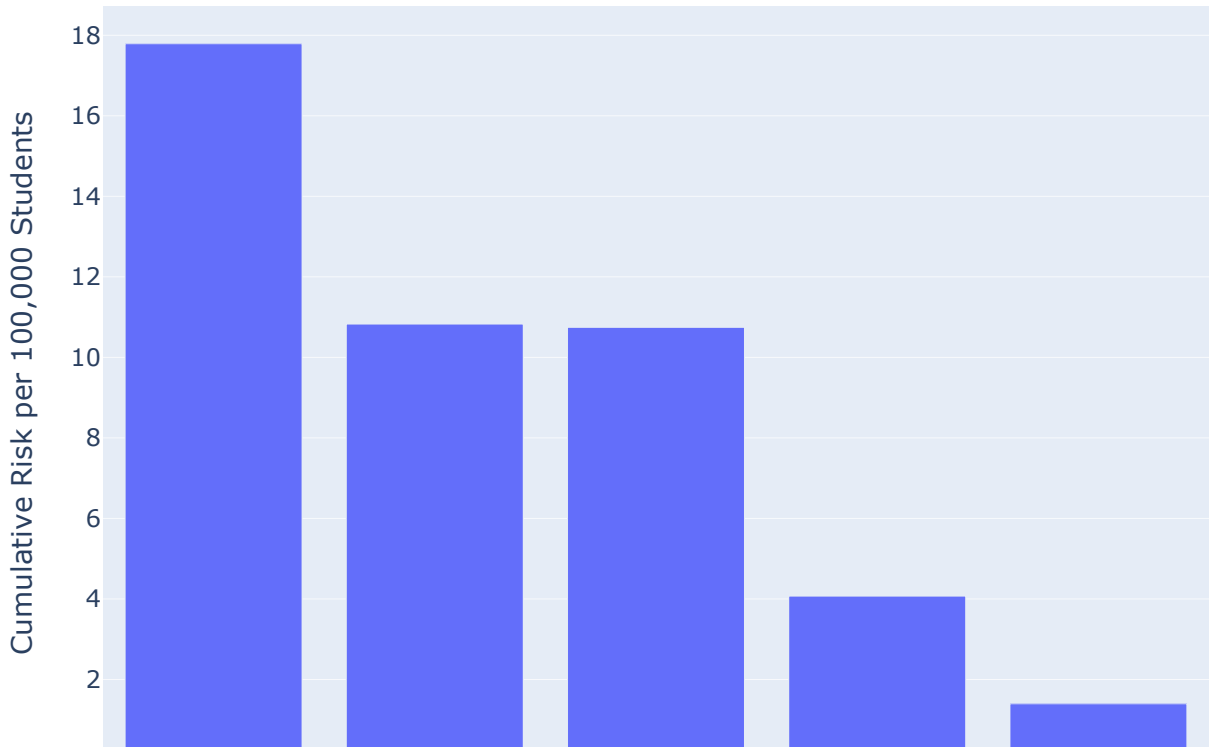
```
import plotly.express as px

dc_annual_enrollment = (
    merged_df_table.loc[
        (merged_df_table["State"] == "DC") &
        (merged_df_table["Year"].between(2018, 2025)),
        ["Year", "total_students"],
    ]
    .copy()
)

dc_situation = (
    incident_df_from_1987.loc[
        (incident_df_from_1987["State"] == "DC") &
        (incident_df_from_1987["Year"].between(2018, 2025)),
        ["Incident_ID", "Year", "Situation"],
    ]
    .copy()
)

dc_situation["Situation"] = (
    dc_situation["Situation"]
    .fillna("Unknown")
    .astype(str)
    .str.strip()
    .replace("", "Unknown")
)

dc_situation_year = (
    dc_situation
    .groupby(["Year", "Situation"], as_index=False)
    .agg(incident_count=("Incident_ID", "size"))
)

top_situations = (
    dc_situation_year
    .groupby("Situation", as_index=False)["incident_count"]
    .sum()
    .sort_values("incident_count", ascending=False)
    .head(6)["Situation"]
    .tolist()
)

dc_situation_year["Situation"] = np.where(
    dc_situation_year["Situation"].isin(top_situations),
    dc_situation_year["Situation"],
    "Other",
)

dc_situation_year = (
    dc_situation_year
    .groupby(["Year", "Situation"], as_index=False)["incident_count"]
    .sum()
    .merge(dc_annual_enrollment, on="Year", how="left")
)

dc_situation_year["risk_per_100k"] = np.where(
    dc_situation_year["total_students"] > 0,
    (dc_situation_year["incident_count"] / dc_situation_year["total_students"]) *
100000,
    np.nan,
)
```

```
situation_order = (
    dc_situation_year
    .groupby("Situation", as_index=False)["risk_per_100k"]
    .sum()
    .sort_values("risk_per_100k", ascending=False)["Situation"]
    .tolist()
)

dc_situation_year["Situation"] = pd.Categorical(
    dc_situation_year["Situation"],
    categories=situation_order,
    ordered=True,
)
dc_situation_year = dc_situation_year.sort_values(["Year", "Situation"])

fig = px.bar(
    dc_situation_year,
    x="Year",
    y="risk_per_100k",
    color="Situation",
    barmode="stack",
    title="DC Annual Risk per 100k by Situation (2018–2025)",
    labels={
        "Year": "Year",
        "risk_per_100k": "Risk per 100,000 Students",
        "Situation": "Situation",
    },
)

fig.update_xaxes(tickmode="array",
tickvals=sorted(dc_situation_year["Year"].unique().tolist()))
fig.update_yaxes(range=[0, merged_df_table["risk_per_100k"].max()])
fig.update_layout(xaxis_title="Year", yaxis_title="Risk per 100,000 Students")
fig.show()

dc_situation_summary = (
    dc_situation_year
    .groupby("Situation", as_index=False)["risk_per_100k"]
    .sum()
    .sort_values("risk_per_100k", ascending=False)
)

fig = px.bar(
    dc_situation_summary,
    x="Situation",
    y="risk_per_100k",
    title="DC Total Risk Contribution by Situation (2018–2025)",
    labels={
        "Situation": "Situation",
        "risk_per_100k": "Cumulative Risk per 100,000 Students",
    },
)

fig.update_layout(xaxis_title="Situation", yaxis_title="Cumulative Risk per 100,000
Students")
fig.show()
```

DC Annual Risk per 100k by Situation (2018–2025)



DC Total Risk Contribution by Situation (2018–2025)

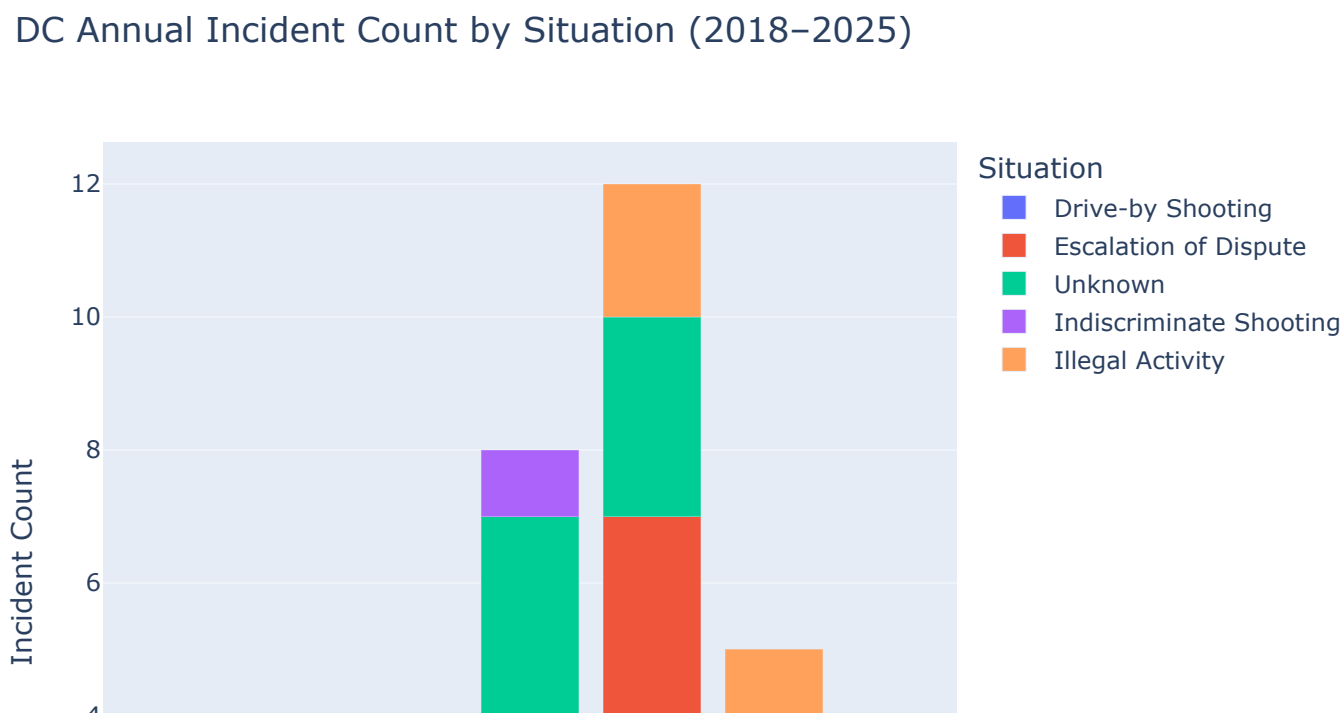


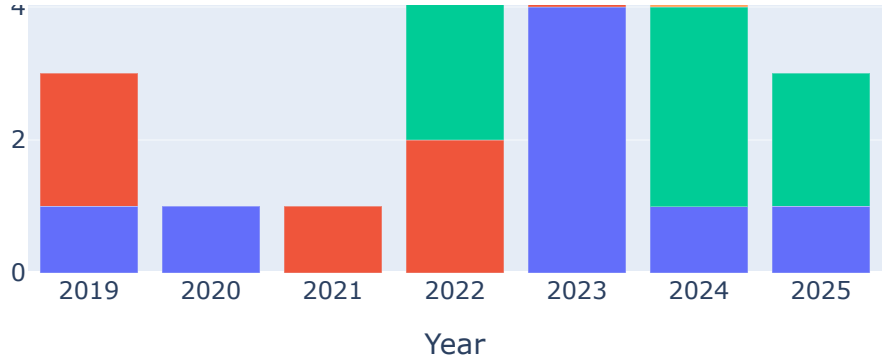
0
Unknown
Escalation of Dispute
Drive-by Shooting
Illegal Activity
Indiscriminate Shooting
Situation

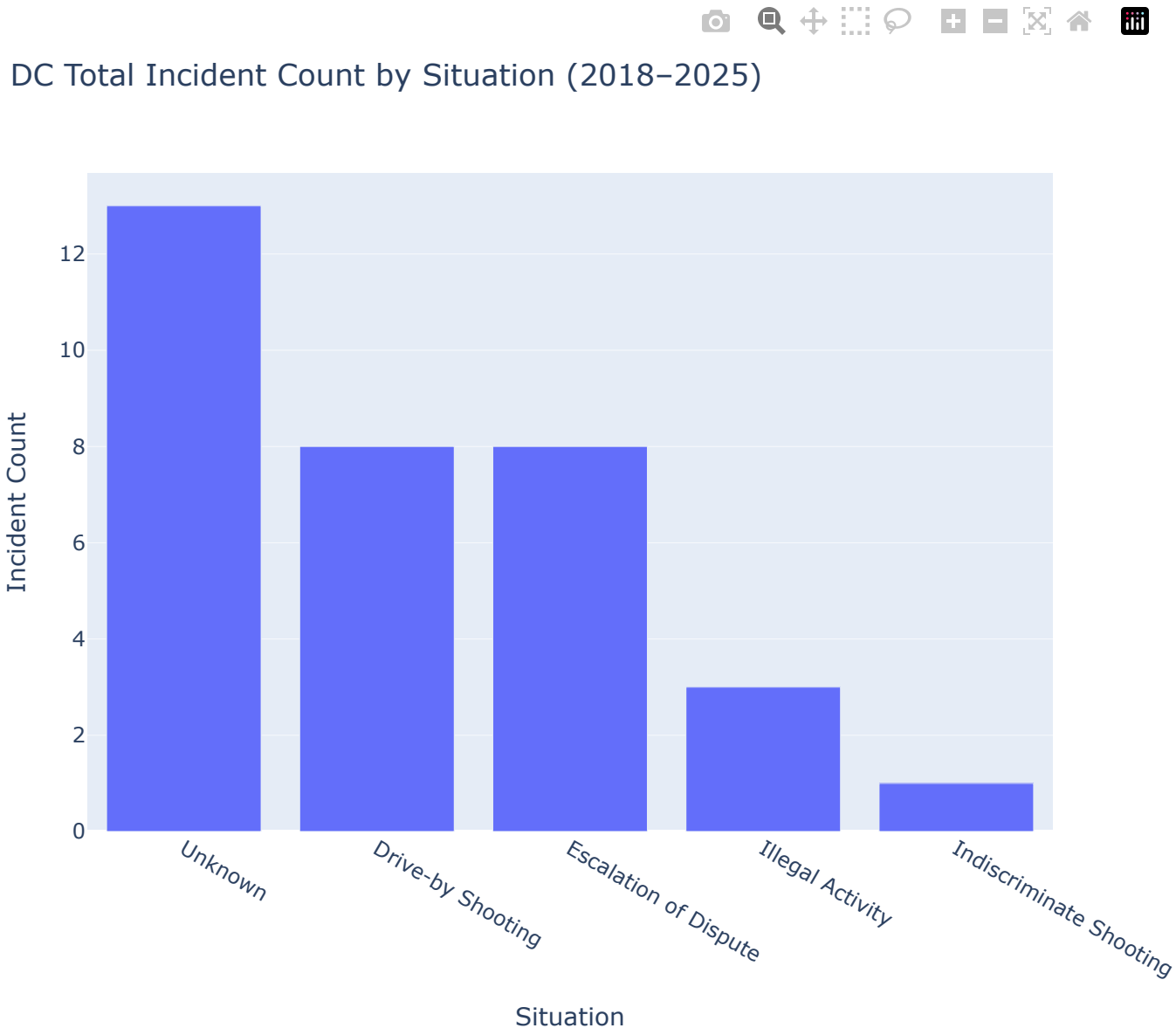
In [20]

```
import plotly.express as px

dc_situation_count = (
    incident_df_from_1987.loc[
        (incident_df_from_1987["State"] == "DC") &
        (incident_df_from_1987["Year"].between(2018, 2025)),
        ["Incident_ID", "Year", "Situation"],
    ]
    .copy()
)

dc_situation_count["Situation"] = (
    dc_situation_count["Situation"]
    .fillna("Unknown")
    .astype(str)
    .str.strip()
    .replace("", "Unknown")
)

dc_situation_count_year = (
    dc_situation_count
    .groupby(["Year", "Situation"], as_index=False)
    .agg(incident_count=("Incident_ID", "size"))
)

top_situations_count = (
    dc_situation_count_year
    .groupby("Situation", as_index=False)["incident_count"]
    .sum()
    .sort_values("incident_count", ascending=False)
    .head(6)["Situation"]
    .tolist()
)

dc_situation_count_year["Situation"] = np.where(
    dc_situation_count_year["Situation"].isin(top_situations_count),
    dc_situation_count_year["Situation"],
    "Other",
)

dc_situation_count_year = (
    dc_situation_count_year
    .groupby(["Year", "Situation"], as_index=False)["incident_count"]
    .sum()
)

situation_count_order = (
    dc_situation_count_year
    .groupby("Situation", as_index=False)["incident_count"]
    .sum()
    .sort_values("incident_count", ascending=False)["Situation"]
    .tolist()
)

dc_situation_count_year["Situation"] = pd.Categorical(
    dc_situation_count_year["Situation"],
    categories=situation_count_order,
    ordered=True,
)
dc_situation_count_year = dc_situation_count_year.sort_values(["Year", "Situation"])

fig = px.bar(
```

```
    dc_situation_count_year,
    x="Year",
    y="incident_count",
    color="Situation",
    barmode="stack",
    title="DC Annual Incident Count by Situation (2018–2025)",
    labels={
        "Year": "Year",
        "incident_count": "Incident Count",
        "Situation": "Situation",
    },
)

fig.update_xaxes(tickmode="array",
tickvals=sorted(dc_situation_count_year["Year"].unique().tolist()))
fig.update_layout(xaxis_title="Year", yaxis_title="Incident Count")
fig.show()

dc_situation_count_summary = (
    dc_situation_count_year
    .groupby("Situation", as_index=False)["incident_count"]
    .sum()
    .sort_values("incident_count", ascending=False)
)

fig = px.bar(
    dc_situation_count_summary,
    x="Situation",
    y="incident_count",
    title="DC Total Incident Count by Situation (2018–2025)",
    labels={
        "Situation": "Situation",
        "incident_count": "Incident Count",
    },
)

fig.update_layout(xaxis_title="Situation", yaxis_title="Incident Count")
fig.show()
```




4
2
0
2019
2020
2021
2022
2023
2024
2025
Year


DC Total Incident Count by Situation (2018–2025)
Incident Count
12
10
8
6
4
2
0
Unknown
Drive-by Shooting
Escalation of Dispute
Illegal Activity
Indiscriminate Shooting
Situation